

Corpus Priming in Polish-to-Interslavic Translation with Large Language Models: A Controlled Study Across Source Regimes

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Abstract

Interslavic (ISV) is a constructed Slavic bridge language with limited digital resources relative to major national Slavic languages. Large language models (LLMs) can be prompted to translate into ISV, but it remains unclear whether providing authentic ISV corpus material as context—*corpus priming*—systematically shifts translation behaviour, and whether any such shift depends on thematic overlap between the source text and the priming corpus. We report a controlled descriptive study of Polish-to-ISV translation under two conditions (Direct vs. Primed) across seven vendor model configurations, three independent repeats per cell, and three source regimes: HIGH (narrative, intentionally higher corpus overlap), LOW (narrative, lower overlap), and UNSEEN (shorter technical/expository voice-biophysics text from a domain absent from the priming corpus). The primary metric is canonical ISV lexical coverage; the paired effect is $\Delta = \text{Primed} - \text{Direct}$. Analysis is descriptive ($n=3$); no significance tests and no causal claims are made. On a matched six-configuration comparison (excluding one invalid LOW pairing), mean Δ is +7.39 percentage points (pp) on HIGH, +8.53 pp on LOW, and +5.78 pp on UNSEEN. Positive mean Δ is common but not universal: two Gemini configurations are negative on HIGH yet positive on LOW and UNSEEN. Qualitative audits show that large HIGH gains often co-occur with corpus-opening motif reuse, whereas LOW and UNSEEN positives frequently involve lexical borrowing and orthographic cleanup without winter-narrative opening near-copy. One Qwen LOW paired cell is invalid due to a Direct/Primed model-identity mismatch and is excluded from primary paired inference. We conclude that authentic ISV corpus priming is associated with generally positive but strongly configuration-dependent coverage shifts that can persist under lower overlap and domain-unseen sources, without establishing that priming universally improves translation quality.

Keywords: Interslavic; machine translation; large language models; in-context learning; low-resource languages

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1 Introduction

Machine translation (MT) into under-resourced target languages remains difficult even as neural and large-language-model (LLM) systems improve for high-resource pairs [Koehn and Knowles, 2017, Maillard et al., 2023]. Interslavic (ISV; also Medžuslovjansky) is a constructed Slavic auxiliary language designed for mutual intelligibility across Slavic speech communities. Relative to national Slavic languages with large web corpora and parallel datasets, ISV occupies a low-resource niche: usable dictionaries and community materials exist, but parallel Polish–ISV training data suitable for conventional neural MT are scarce, and automated evaluation infrastructure is not standardized in the same way as for WMT high-resource directions.

For Slavic speakers, Polish-to-ISV translation is practically interesting because Polish is a high-resource national language while ISV is intended as a bridge form. For NLP methodology, the setting is interesting because it combines (i) a target language with limited bitext, (ii) available monolingual or curated target-side resources, and (iii) commercial LLMs that can be prompted without gradient updates. The methodological risk is equally clear: an attractive prompting trick can be mistaken for a validated translation method if experiments do not separate instruction-only behaviour from corpus-conditioned behaviour, and if they do not test whether apparent gains depend on thematic overlap between the source and the supplied corpus.

Prompting LLMs with instructions and in-context examples has become a practical route to translation without fine-tuning [Brown et al., 2020, Hendy et al., 2023, Jiao et al., 2023]. A natural engineering idea for ISV is to place authentic ISV text in the prompt—here called *corpus priming*—so that the model can condition generation on attested target-language material rather than on an instruction alone. That idea is appealing, but it is not self-evidently beneficial. Corpus material may encourage useful lexical and orthographic alignment; it may also encourage motif copying, register drift, or metric gains that do not correspond to communicative quality. Moreover, if a source text already overlaps thematically with the priming corpus, apparent “priming gains” may partly reflect overlap rather than a generalizable priming intervention.

This paper reports EXP-004 Phase 2B: a controlled Polish-to-ISV study that separates Direct translation (no corpus) from Primed translation (authentic ISV corpus first, then translation), using identical model configurations, three independent repeats, and three source regimes that vary corpus overlap and genre:

- **HIGH**: narrative fiction deliberately inspired by the priming corpus;
- **LOW**: narrative fiction with intentionally weak thematic overlap;
- **UNSEEN**: shorter technical/expository educational text on human-voice biophysics, from a domain absent from the priming corpus registers.

The primary quantitative quantity is the paired difference in *canonical ISV coverage*, $\Delta = \text{Primed} - \text{Direct}$. Because each configuration cell has $n=3$ repeats, all quantitative reporting is descriptive: we do not perform significance testing and we do not claim causal identification of priming as a mechanism.

The study is deliberately comparative across regimes rather than optimized for a single leaderboard score. **HIGH** asks whether priming is associated with coverage shifts when overlap is intentionally high. **LOW** asks whether positive shifts can still appear when overlap is intentionally reduced. **UNSEEN**

asks whether positive shifts appear for a short technical/expository domain that is not represented in the narrative priming materials. Together, the three regimes allow a restrained statement about persistence and heterogeneity without pretending that overlap, length, and genre have been orthogonally crossed.

Contributions. Conservatively stated, this paper contributes:

1. a reproducible Direct/Primed protocol for Polish-to-ISV LLM translation across seven configurations and three source regimes;
2. a descriptive cross-regime evidence record for canonical coverage Δ , including matched-configuration summaries and one documented invalid pairing cell;
3. a qualitative paired audit synthesizing corpus-opening reuse, lexical borrowing, and orthographic change without equating metric gains with translation quality.

We do *not* claim that corpus priming universally improves translation, that higher overlap causes larger effects, or that any configuration is globally best.

Scope of claims. Throughout the paper we use associative language (“associated with”, “observed”, “descriptive”) rather than causal language (“improves”, “causes”, “proves”), except where we explicitly reject causal slogans. This is not hedging for its own sake; it is required by the $n=3$ descriptive design and by the confounded regime factors.

2 Related Work

LLM-based machine translation. LLMs can perform translation in zero- and few-shot settings without task-specific fine-tuning [Brown et al., 2020]. Systematic evaluations report competitive quality for high-resource directions and weaker behaviour for low-resource pairs, with sensitivity to prompting strategy and domain [Hendy et al., 2023, Jiao et al., 2023, Zhu et al., 2024]. Document-level and domain-shift analyses further show that LLM translation behaviour is not a single scalar property of a model name: prompt packaging and domain can change outcomes [Hendy et al., 2023]. Comparisons among GPT-family systems also illustrate that engine choice can dominate superficial “ChatGPT vs. translator” contrasts [Jiao et al., 2023]. These findings motivate careful experimental design when the target language is under-resourced: gains observed under favourable prompting may not transfer across domains or source types.

A recurring methodological lesson in this literature is that prompting is itself an experimental factor. Changing the presence, length, or identity of in-context material can move measured translation scores even when the underlying model family is held constant. Our Direct vs. Primed contrast is intentionally narrow: it varies corpus presence while freezing the authentic corpus bytes and the source text within each regime.

Low-resource and multilingual MT. Data scarcity, domain mismatch, and rare-word behaviour remain core difficulties for neural MT [Koehn and Knowles, 2017]. Recent work continues to study how minimal or carefully selected data can still yield useful systems [Maillard et al., 2023]. Low-resource MT research often focuses on parallel-data collection, transfer learning, or multilingual training objectives. Our setting

differs: we do not train an NMT model on Polish–ISV bitext. Instead we evaluate commercial/API LLM configurations under a fixed Direct vs. Primed prompting contrast. This makes the work closer to an intervention study on prompting context than to a bitext-training bake-off. The practical motivation remains related: when bitext is scarce, target-side authentic text may be the most available resource for conditioning generation.

In-context learning and example-based contextualization. Few-shot in-context learning [Brown et al., 2020] shows that providing demonstrations in the prompt can change task behaviour without weight updates. Adaptive or example-conditioned LLM translation explores related ideas in which target-side or parallel material is supplied to guide generation [Moslem et al., 2023]. Multilingual LLM-MT analyses likewise emphasize that in-context examples and language coverage interact [Zhu et al., 2024]. Our *corpus priming* condition supplies a frozen authentic ISV multi-register corpus before the translation request. It is therefore related to in-context conditioning, but it is not a classical k -shot bitext demonstration protocol, nor a fine-tuning method. The distinction matters for interpretation: a positive Δ indicates an association between corpus-conditioned prompting and the coverage metric, not proof that the model acquired a new grammar from the corpus.

Evaluation of translation and low-resource systems. Surface n -gram metrics such as BLEU [Papineni et al., 2002] and learned metrics such as COMET [Rei et al., 2020] are widely used for MT evaluation; LLMs have also been studied as quality estimators [Kocmi and Federmann, 2023]. Each family of metrics encodes assumptions about references, human ratings, or learned representations that may not hold for Interslavic. For ISV, we do not treat BLEU/COMET as primary endpoints in this experiment. Phase 2B uses a project-specific resource-supported coverage stack (canonical coverage, broader coverage, unresolved rate, and orthography outside-inventory), designed around available ISV lexical/orthographic resources. We explicitly warn that coverage is not a complete proxy for translation quality (Section 7). In low-resource settings, mismatched evaluation choices can create false confidence; we therefore keep metric language operational and separate qualitative audits from coverage tables.

Interslavic language technology. Interslavic is a community-developed Slavic auxiliary language with published grammars, dictionaries, and educational materials. Computational work on Slavic NLP is extensive for national languages, but peer-reviewed evaluations of Polish-to-ISV LLM translation under controlled corpus priming remain scarce. We therefore treat ISV primarily through operational definitions in this repository experiment rather than through a large prior MT literature. Additional language-description citations suitable for a camera-ready related-work paragraph are listed in `REFERENCES_NEEDED.md` when full bibliographic verification is still pending.

Positioning of this study. Relative to LLM-MT evaluations that sweep many language pairs, this paper is narrow and deep: one source language, one constructed target, three source regimes, seven configurations, and paired repeats. Relative to purely anecdotal prompting notes, it is comparatively strict: frozen hashes, integrity gates, invalid-cell documentation, and an evidence map from claims to artifacts. That trade-off is intentional for an under-resourced target where over-claiming would be especially costly.

3 Research Questions

We organize the study around six empirical questions. They are answered descriptively from the Phase 2B evidence pack; none is treated as rhetorically settled in advance.

- RQ1** Does authentic ISV corpus priming produce a measurable shift in Polish-to-ISV translation outputs under the canonical coverage metric?
- RQ2** Does this shift persist when the source text has lower thematic overlap with the priming corpus (LOW)?
- RQ3** Does a positive paired shift persist for a domain-unseen, technical/expository source (UNSEEN)?
- RQ4** How heterogeneous are priming effects across the seven model configurations?
- RQ5** What qualitative textual changes accompany positive or negative shifts in canonical coverage?
- RQ6** Which observed changes appear related to corpus-opening copying, lexical adoption, or orthographic normalization?

RQ1–RQ3 address magnitude and direction of Δ across regimes. RQ4 addresses configuration dependence. RQ5–RQ6 connect metric movement to audited textual behaviour without asserting causal mechanisms.

The RQs are ordered from measurement (RQ1) through generalization across sources (RQ2–RQ3) to heterogeneity and mechanism-facing description (RQ4–RQ6). This order mirrors the analysis workflow actually used in the repository: regime aggregates first, then combined synthesis, then qualitative audits.

4 Experimental Design

4.1 Conditions

Each translation session is assigned to one of two conditions:

- **Direct:** the model receives the Polish source and a translation instruction into Interslavic, without the authentic priming corpus.
- **Primed:** the model first receives the frozen authentic ISV corpus, then the translation instruction and Polish source.

The corpus is never shortened for any configuration. For Gemini Primed runs, corpus delivery uses a recorded multi-message protocol (corpus in message 1; translation request in message 2) because of interface/context-window constraints (Section 12). All other scientific factors—source text, configuration identity, and repeat structure—are held as constant as the collection interfaces allow.

Table 1: Experimental design and source regimes (Phase 2B).

Regime	Source (frozen v1)	Design notes
HIGH	<i>Iskra i Wieloryb...</i> (30 061 B)	Narrative; intentionally higher corpus overlap
LOW	<i>Podkłady</i> (15 249 B)	Narrative; intentionally lower overlap
UNSEEN	<i>Ćwiczenie 2.2 — Biofizyka głosu...</i> (5 676 B)	Expository technical; domain unseen

Conditions: Direct vs. Primed. Configurations: 7. Repeats: 3 independent sessions. Planned runs per regime: 42. Effect: Δ = Primed – Direct (canonical coverage, descriptive). Priming corpus: phase2a-authentic-isv v1 (SHA-256 aaad28e4 . . . a857), never shortened.

4.2 Pairing and effect definition

For each configuration and replicate $i \in \{r01, r02, r03\}$,

$$\Delta_i = C(\text{Primed}_i) - C(\text{Direct}_i),$$

where C denotes canonical ISV coverage (reported as a percentage; Δ in percentage points). Configuration-level summaries use the mean of the three replicate Δ_i values. All comparisons are descriptive. The paired construction is the reason a Direct/Primed model-identity mismatch invalidates a cell: Δ is not defined as a same-model priming effect unless both sides share the intended configuration.

4.3 Factorial structure

Within each source regime the planned design is:

$$7 \text{ configurations} \times 2 \text{ conditions} \times 3 \text{ repeats} = 42 \text{ runs.}$$

Repeats are independent fresh sessions. The same seven configurations are used in HIGH, LOW, and UNSEEN. Across the full Phase 2B programme this yields $3 \times 42 = 126$ planned runs, of which one LOW paired configuration is invalid for priming inference.

4.4 Why the analysis is descriptive

With $n=3$ per configuration cell, the study is not powered for inferential claims about population effects. We report means, standard deviations, ranges, and sign consistency. We do not compute p -values, confidence intervals intended for inferential generalization, or model rankings. This is a deliberate epistemological choice aligned with the sample size, not an accidental omission.

4.5 Operator protocol notes

Runs were collected as independent sessions following the Phase 2B kits prepared for each regime. Intake verification checked completeness and identity metadata before evaluation. Analysis scripts aggregate already-evaluated runs and do not call LLMs. These process controls do not eliminate vendor stochasticity, but they make the Direct/Primed pairing auditable after the fact.

5 Corpus and Source Regimes

5.1 Authentic priming corpus

All Primed runs use the frozen authentic three-register corpus `phase2a-authentic-isv v1` (SHA-256 `aaad28e43935a40313585d77a33bfc788d97e8d69b081f9486af74d52ca1a857`). The corpus is held constant across regimes and configurations. It is never shortened, paraphrased, or filtered to evade interface constraints. Prior corpus self-evaluation under the same evaluator stack provides a reference ceiling for reading coverage magnitudes; it is not a translation baseline and is not used as a Direct/Primed control.

Keeping the corpus fixed is essential for the scientific contrast: regime differences are differences in the Polish source (and associated genre/length/overlap properties), not silent edits to the priming material.

5.2 Source regimes

Table 1 summarizes the three regimes. Length and genre differ jointly with overlap by design.

HIGH. Frozen Polish narrative *Iskra i Wieloryb — wersja z oryginalnymi nazwami* v1 (SHA-256 `ab8a0dcf7352789c09c4aca132c086999c861407e4cd682ee9414aab5b792f63`; 30 061 bytes / 440 lines). Classification: `high_overlap_corpus_inspired`. The story deliberately shares motifs with the priming corpus (for example winter narrative material associated with the corpus register excerpt used in qualitative opening checks). HIGH is therefore not an “independent same-topic” control; it is the high-overlap arm of the design.

LOW. Frozen Polish narrative *Podkłady* v1 (SHA-256 `ce1c4fca03fe9cb2c5f8181ab45c91767759a0c785f0` 15 249 bytes / 141 lines). Classification: `low_overlap`. Contemporary municipal/archive plot with intentionally weak thematic overlap with the priming corpus and with the HIGH story. Clean-prose boundary decisions (exclusion of casting notes and unrelated preamble material) were operator-approved before freeze.

UNSEEN. Frozen Polish educational theory text *Ćwiczenie 2.2 — Biofizyka głosu ludzkiego* v1 (SHA-256 `cd3bfb9a819b415e3cfb382e0737ba22540ccb679d0d34983c40dfbf89a9f7f4`; 5 676 bytes after validation hash basis). Classification: `unseen_domain`. Voice/speech biophysics exposition; domain absent from HIGH/LOW narrative registers and from the priming corpus domains under the project’s validation record. Exact bibliographic provenance of the teaching-derived source is incompletely known and is disclosed as a limitation.

Design confounds (disclosed). Overlap, length, and genre are not orthogonally crossed. UNSEEN is shorter and expository; HIGH/LOW are longer narratives. Magnitude comparisons across regimes are therefore descriptive and must not be read as causal attributions to overlap alone. The manuscript treats these differences as first-class design characteristics rather than as nuisances to be narrated away after seeing the results.

Table 2: Model configurations evaluated in all three regimes (canonical labels).

Short label	Canonical configuration name
Gemini ON	Gemini 3.6 Flash — extended thinking ON
Gemini OFF	Gemini 3.6 Flash — extended thinking OFF
Claude Sonnet 5	Claude Sonnet 5 — Medium (default)
DeepSeek Expert ON	DeepSeek V3 Expert — DeepThink ON
Qwen 3.8 Max Fast	Qwen 3.8 Max — Fast
GPT-5.6 Luna	GPT-5.6 Luna — thinking OFF
Grok 4.5 Fast	Grok 4.5 Fast

5.3 Why three regimes rather than one

A single-regime study could not separate high-overlap narrative behaviour from lower-overlap or domain-unseen behaviour. Three regimes increase confounds (length/genre) but buy interpretive breadth: persistence of positive Δ under LOW and UNSEEN cannot be dismissed as merely HIGH-overlap artefact, while HIGH still documents that negatives are possible even when overlap is intentional.

6 Models and Experimental Conditions

We evaluate seven Phase 2B configurations, named exactly as in the canonical experiment artifacts (Table 2). Configurations differ by vendor model and, where applicable, thinking/mode switches exposed by the interface used for collection. No configuration is treated as a gold standard.

Gemini ON vs. OFF. Two Gemini 3.6 Flash configurations differ only in the extended-thinking switch (ON vs. OFF). Both use the multi-message Primed protocol described in Section 12.

What is held constant. Within a regime, all configurations translate the same frozen Polish source under the same Direct/Primed definitions, with three independent repeats. Cross-regime comparisons reuse the same configuration list.

What is not claimed. Vendor display names and mode switches are operational labels for reproducibility. We do not equate interface labels with internal model checkpoints beyond what the operator interfaces exposed at collection time, and we do not rank configurations by quality.

6.1 Collection interfaces as part of the method

Because the experiment uses vendor chat/API surfaces rather than a single open-weights checkpoint under identical decoding controls, configuration identity includes the operator-facing mode switches recorded at collection time. That is a feature of the ecological validity of the study and a limitation for mechanistic attribution to a specific open checkpoint. The manuscript therefore reports configuration labels exactly and avoids rewriting them into unofficial synonyms.

6.2 Non-goals of the configuration set

The seven configurations were chosen as representative Phase 2B operator-facing setups, not as an exhaustive survey of all vendors or decoding hyperparameters. Absence of an open-weights local baseline is a limitation for mechanistic replication, not an accidental omission in the Results tables.

7 Evaluation Methodology

7.1 Primary and secondary metrics

Each completed run is scored with the project’s deterministic ISV evaluation stack (no LLM calls during scoring). The metrics used in Phase 2B are:

- **canonical coverage** (primary): resource-supported coverage against the canonical ISV lexical inventory used by the evaluator;
- **broader resource-supported coverage**: a broader inventory variant;
- **unresolved rate**: proportion of tokens not resolved to supported inventory entries under the evaluator construction;
- **orthography outside-inventory**: character-level count of orthographic material outside the expected inventory (separate diagnostic).

Manuscript quantitative claims focus on canonical coverage and its paired Δ . Other metrics are retained in the machine-readable datasets for audit and are not collapsed into a single composite score. In particular, unresolved rate tends to move inversely with canonical coverage under the evaluator construction; reporting both without inventing a composite avoids double-counting the same inventory phenomenon under two names.

7.2 What coverage is not

Canonical coverage is a screening metric for resource-supported ISV lexical realization. It is *not* claimed to be equivalent to:

- human-judged translation quality,
- adequacy/fluency in the sense of DA/MQM protocols,
- BLEU [Papineni et al., 2002] or COMET [Rei et al., 2020].

Orthographic cleanup or substitution of inventory-supported synonyms can move coverage without implying a global quality improvement. Conversely, communicative improvements that do not increase inventory hits may be invisible to the metric. This asymmetry is why the paper pairs every major quantitative claim with qualitative audits and explicit limitations.

7.3 Paired analysis unit

The experimental unit for priming effects is the Direct $\leftrightarrow$ Primed pair within a configuration and replicate, not an unpaired mixture of runs. Configuration summaries average three paired Δ_i values. Invalid pairs are excluded from primary paired inference rather than imputed.

7.4 Qualitative audit

For each regime, all valid Direct $\leftrightarrow$ Primed pairs were inspected in a descriptive qualitative audit (HIGH and UNSEEN: 21 pairs; LOW primary: 18 valid pairs). Audits record apparent classes such as lexical borrowing, orthographic cleanup, corpus-opening alignment, and degradations, without adjudicating full linguistic correctness. Opening-overlap checks against a corpus narrative excerpt are used as an operational detector for winter-opening near-copy, especially to contrast HIGH against LOW/UNSEEN.

7.5 Reporting conventions used in Results

Unless otherwise stated, coverage means are percentages averaged over three Direct or three Primed runs, and Δ means are averages of three paired differences. We report two decimal places in percentage points when that precision is present in the combined tables. Sign consistency is reported as all-positive, all-negative, or mixed within the three replicates. Invalid cells are marked explicitly rather than imputed.

8 Results

All quantitative results below reconstruct from the canonical combined synthesis and the underlying HIGH/LOW/UNSEEN analysis artifacts. No numbers were re-estimated for this manuscript. Where we round to two decimal places in percentage points, rounding follows the combined evidence tables and does not change sign or qualitative interpretation.

8.1 Integrity and validity

HIGH and UNSEEN each collected and evaluated 42/42 planned runs with seven valid paired configurations (21 pairs). LOW collected 42/42 retained runs but has six valid paired configurations (18 pairs) because the Qwen cell is `INVALID -- MODEL MISMATCH` (Section 12). Integrity gates (unique run IDs; complete Direct $\leftrightarrow$ Primed pairing structure) pass on all three regimes. These integrity facts are prerequisites for interpreting any mean Δ : a missing or mismatched pair would silently break the Direct/Primed contrast.

8.2 Central configuration-level Δ table

Table 3 is the primary quantitative table for later discussion and for RQ1–RQ4. Reading the table by columns emphasizes regime behaviour; reading by rows emphasizes configuration heterogeneity. Both views are necessary. A column-only summary would hide Gemini’s HIGH negatives; a row-only focus on one “successful” configuration would hide attenuation and invalid cells elsewhere.

Table 3: Central cross-regime configuration mean Δ (canonical coverage, percentage points). LOW Qwen cell is invalid and shown as —.

Configuration	HIGH Δ	LOW Δ	UNSEEN Δ
Gemini ON	−0.55	+13.23	+5.68
Gemini OFF	−1.75	+10.40	+6.09
Claude Sonnet 5	+14.28	+12.11	+6.42
DeepSeek Expert ON	+9.25	+6.19	+3.83
Qwen 3.8 Max Fast	+5.18	—	+3.52
GPT-5.6 Luna	+8.62	+5.31	+5.52
Grok 4.5 Fast	+14.49	+3.95	+7.12

Source:

experiments/exp004-modelscreen/phase2b/analysis/combined/cross_regime_table.csv. Each cell is the mean of three replicate Δ_i values except LOW Qwen (invalid model mismatch).

Table 4: Direct and Primed mean canonical coverage (%) by configuration and regime. LOW Qwen paired means are omitted (invalid).

Config.	HIGH		LOW		UNSEEN	
	D	P	D	P	D	P
Gemini ON	68.68	68.13	64.33	77.56	65.77	71.45
Gemini OFF	67.20	65.45	64.51	74.92	63.85	69.95
Claude	69.29	83.57	65.72	77.83	66.07	72.48
DeepSeek	74.53	83.78	68.93	75.12	67.97	71.80
Qwen	76.36	81.54	—	—	67.65	71.17
GPT-5.6	73.54	82.16	70.75	76.06	68.53	74.05
Grok	68.35	82.84	69.32	73.27	65.90	73.02

Source: analysis/combined/analysis.md canonical cross-regime table.

8.3 Direct and Primed means

In addition to Δ , Table 4 reports configuration-level Direct and Primed mean coverage. This table clarifies whether a positive Δ arises from a large Primed gain, a low Direct baseline, or both. For example, Gemini LOW combines comparatively low Direct means with substantially higher Primed means, whereas several HIGH positives combine mid Direct means with Primed means above 80%.

8.4 All-configuration vs. matched-six summaries

Unequal all-configuration means must not be treated as a matched three-regime average:

- HIGH: mean of 7 configuration Δ s = +7.07 pp (5 positive, 2 negative);
- LOW: mean of 6 valid configuration Δ s = +8.53 pp (6 positive);
- UNSEEN: mean of 7 configuration Δ s = +5.45 pp (7 positive).

For direct cross-regime comparison we use the matched six configurations available in all regimes (Table 5): HIGH +7.39 pp, LOW +8.53 pp, UNSEEN +5.78 pp. The matched-six construction excludes Qwen because LOW cannot supply a valid same-model paired effect for that configuration. This exclusion is a validity decision, not a performance ranking.

Table 5: Matched-six regime summary (identical six configurations on HIGH/LOW/UNSEEN; Qwen excluded).

Regime	Mean Δ (pp)	n configs
HIGH	+7.39	6
LOW	+8.53	6
UNSEEN	+5.78	6

Included: Gemini ON, Gemini OFF, Claude Sonnet 5, DeepSeek Expert ON, GPT-5.6 Luna, Grok 4.5 Fast. Descriptive only; not a statistical test.

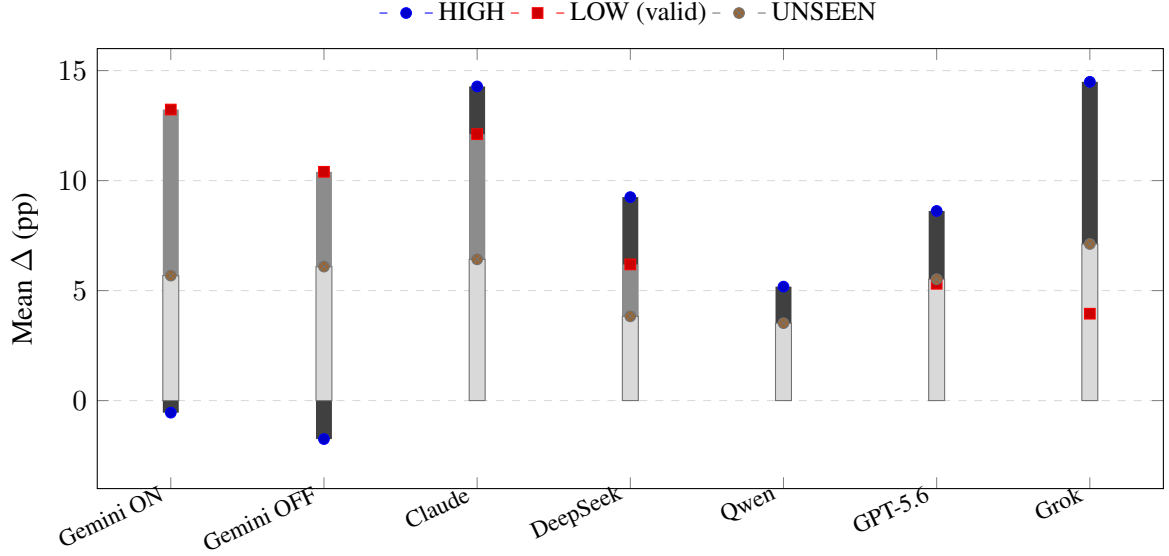


Figure 1: Configuration-level mean canonical-coverage Δ across regimes. LOW omits Qwen (invalid Direct/Primed model mismatch). Descriptive only; no inferential intervals.

8.5 Replicate consistency

Sign consistency of the three replicate Δ_i values varies by configuration and regime (Figure 2):

- HIGH: Gemini ON/OFF and Qwen show mixed signs; Claude, DeepSeek, GPT, and Grok are all-positive.
- LOW (valid): five configurations are all-positive; Grok is mixed (one negative replicate).
- UNSEEN: all 21 replicate Δ s are positive.

Particularly variable HIGH cells include Grok ($SD(\Delta) \approx 5.90$ pp) and Qwen ($SD(\Delta) \approx 5.56$ pp). More stable examples include GPT on HIGH ($SD(\Delta) \approx 1.34$ pp) and Claude/DeepSeek on UNSEEN ($SD(\Delta) \approx 1.11 / 1.06$ pp). Variance is reported as a descriptive property of the sample, not as a formal test of equality of variances.

8.6 Regime-level reading of the metric

On HIGH, mean Δ spans a wide range: from -1.75 pp (Gemini OFF) to $+14.49$ pp (Grok). On LOW, valid means are all positive and Gemini becomes strongly positive. On UNSEEN, means are moderate

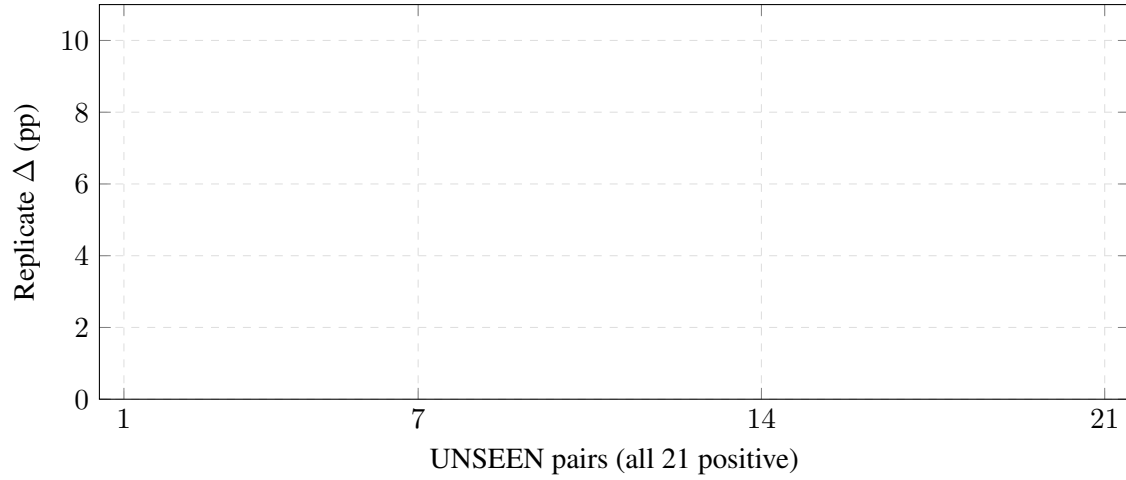


Figure 2: UNSEEN replicate-level paired Δ values (canonical coverage). All 21 Direct $\leftrightarrow$ Primed pairs are positive. Points are ordered by configuration (Gemini ON, Gemini OFF, Claude, DeepSeek, Qwen, GPT-5.6 Luna, Grok), three replicates each. Descriptive display only.

and uniformly signed at both configuration and replicate levels. These patterns answer different questions: HIGH shows that priming is not uniformly helpful under the metric even when overlap is high; LOW and UNSEEN show that positive associations are not confined to high-overlap narrative material.

8.7 Answering RQ1–RQ3 at the metric level

RQ1: priming is associated with measurable Δ shifts, but not with a single universal sign on HIGH. RQ2: on LOW, every *valid* configuration mean Δ is positive; Gemini means are large (+13.23 / +10.40 pp). RQ3: on UNSEEN, every configuration mean Δ is positive and every replicate Δ is positive. These statements are descriptive coverage observations, not quality or causality claims.

8.8 Configuration-level commentary (descriptive)

The following comments restate Table 3 in prose so that later Discussion can refer to patterns without reintroducing numbers from memory.

Gemini. Both Gemini configurations have negative mean Δ on HIGH (−0.55 and −1.75 pp) with mixed replicate signs. On LOW they become the largest valid positives (+13.23 and +10.40 pp), all replicates positive. On UNSEEN they remain positive (+5.68 and +6.09 pp), all replicates positive. This is the clearest cross-regime reversal in the study.

Claude Sonnet 5. Claude is positive in every regime mean and every replicate in the primary tables. Magnitudes are large on HIGH/LOW and smaller on UNSEEN. The pattern is persistence with attenuation toward the short technical source, not a sign flip.

DeepSeek Expert ON. DeepSeek is likewise persistently positive, with a smooth attenuation sequence +9.25 $\rightarrow$ +6.19 $\rightarrow$ +3.83 pp. Among valid configurations, it is one of the clearest monotonic magnitude declines across the three regimes.

Qwen 3.8 Max Fast. Qwen contributes valid positive means on HIGH (+5.18 pp, mixed replicates) and UNSEEN (+3.52 pp, all positive). LOW remains invalid; therefore Qwen cannot enter matched-six summaries.

GPT-5.6 Luna. GPT is positive everywhere in the primary means, with relatively stable HIGH replicates and similar LOW/UNSEEN magnitudes. It is an example of a moderate, persistent association rather than an extreme HIGH-only spike.

Grok 4.5 Fast. Grok shows a large HIGH mean (+14.49 pp), a much smaller LOW mean (+3.95 pp) with one negative replicate, and a positive UNSEEN mean (+7.12 pp). The scientific point is magnitude instability across regimes, not a claim that Grok is “best” or “worst.”

9 Qualitative Analysis

This section synthesizes the regime qualitative audits and the combined qualitative synthesis. Examples are traceable to audited configuration/replicate cells; no invented illustrations are used. The audit language is intentionally cautious: labels such as “lexical borrowing” or “improvement” are apparent classes relative to orthographic/resource signals and corpus-overlap observations, not adjudicated proof of linguistic correctness.

9.1 Corpus-opening and motif copying

On HIGH, large positive Δ cells often co-occur with primed openings that track the authentic corpus winter narrative (audit cue: *Ljudi govoret, že v tamtoj denj. . .*). Documented examples include Claude HIGH (mean $\Delta + 14.28$ pp) and DeepSeek HIGH (mean $\Delta + 9.25$ pp), where Kit→Velryb and corpus lexis such as *pomalo, jedino, črěz* appear with opening alignment. Qwen HIGH r01–r02 show opening adoption with positive Δ , while r03 is negative despite opening presence—a reminder that opening copy is not sufficient for a positive metric shift.

On LOW, primed openings remain recognizably *Podkłady* (Katarzyna/Katarina + archive). Opening Jaccard against the winter corpus excerpt stays low (maximum ≈ 0.09 in the LOW audit). Nevertheless, Gemini LOW shows large positive mean Δ without systematic winter-opening near-copy. This contrast is one of the strongest qualitative results in the study: the textual signature that accompanies many HIGH positives is largely absent from LOW positives.

On UNSEEN, primed texts open with exercise/biophysics titles (e.g. *Vježba/Upraznjenje + Biofizika . . . glasa*), not the winter story; opening Jaccard remains low for all 21 pairs. If corpus priming “worked” only by transplanting the winter narrative, UNSEEN positives should not occur in this form.

9.2 Lexical borrowing and orthographic cleanup

Across regimes, primed outputs frequently show corpus-register lexical items (*črěz, jedino, govoret, pomalo, no* for Polish *ale*) while retaining source plot or expository structure. Concrete LOW example: Gemini ON r01, $\Delta + 15.84$ pp, with *črěz/govoret* and no winter-opening rewrite. Orthographic / Polish-residue reductions often co-occur with positive Δ (e.g. LOW Claude r01, outside-inventory 301 → 11 with $\Delta + 13.82$ pp; HIGH Claude r01, 365 → 18), but dimensions can diverge (coverage down with

ortho up, or both up). Therefore orthographic cleanup is best described as a recurrent correlate, not as a sufficient explanation of Δ .

9.3 Positive Δ without obvious corpus-opening copying

LOW Gemini ON/OFF cell means (+13.23 / +10.40 pp) and UNSEEN Gemini ON r03 (+8.84 pp) illustrate positive coverage shifts with little or no winter-motif transplant. UNSEEN Qwen r02 (+1.27 pp) shows the complementary pattern: markers may be present without a large coverage move. These cases are central to RQ6: metric-positive priming is compatible with lexical/orthographic calibration rather than wholesale narrative copying, and surface corpus cues are not a reliable thermometer of Δ magnitude.

9.4 Structural and domain preservation

LOW primed outputs keep the urban/archive plot; UNSEEN keeps educational biophysics terminology (*formant**, *rezonan**, *biofizik**) in both Direct and Primed. Title/name churn occurs (HIGH whale-title variants; LOW Katarina/Katarzyna; UNSEEN exercise-title variants) and should not be equated with priming success. Markdown or formatting artifacts can also appear (e.g. UNSEEN GPT primed r03 title formatting) without invalidating run completeness under the project's intake rules; such surface noise is another reason not to equate coverage with polished quality.

9.5 RQ5 summary

Positive coverage shifts are accompanied by heterogeneous textual changes: HIGH often shows opening/motif alignment plus lexical adoption; LOW/UNSEEN more often show lexical and orthographic shifts inside preserved source structure. Negative or mixed HIGH Gemini cells show reshuffles without reliable corpus-opening adoption and sometimes show rising Polish-residue or outside-inventory signals. The qualitative layer therefore supports configuration- and regime-dependent descriptions rather than a single Primed “style.”

10 Cross-Regime Analysis

10.1 Overall direction

Matched-six means are positive in all three regimes (Table 5). UNSEEN adds a uniformly positive replicate pattern (21/21) on a short technical source. LOW shows that large positive Δ can appear under intentionally low thematic overlap. HIGH is the only regime with negative configuration mean Δ (both Gemini conditions). Taken together, the regimes support a tempered summary: positive associations are common across regimes, but HIGH demonstrates that common is not universal at the configuration level.

10.2 Configuration-specific patterns

We characterize heterogeneity without ranking:

- **Gemini ON/OFF:** negative or near-zero mean Δ on HIGH with mixed replicates; strongly positive on LOW; moderately positive on UNSEEN. This is a clear HIGH→LOW direction reversal.

- **Claude Sonnet 5:** positive mean Δ in all three regimes; HIGH/LOW large; UNSEEN smaller (+6.42 pp).
- **DeepSeek Expert ON:** persistently positive with attenuation HIGH→LOW→UNSEEN (+9.25 → +6.19 → +3.83 pp).
- **GPT-5.6 Luna:** persistently positive; HIGH larger than LOW/UNSEEN, which are similar (+5.31 / +5.52 pp).
- **Grok 4.5 Fast:** large positive HIGH; much smaller LOW mean with one negative replicate; positive UNSEEN.
- **Qwen 3.8 Max Fast:** positive HIGH and UNSEEN means; no valid LOW paired effect.

These patterns differ in kind, not merely in rank order. Reversal (Gemini), attenuation (DeepSeek), magnitude instability without mean-sign flip (Grok), and missing cells (Qwen LOW) are distinct descriptive phenomena.

10.3 Overlap is not a sufficient explanation

If larger overlap were necessary for positive priming Δ , LOW and UNSEEN should not show widespread positives. They do. Gemini’s largest positives appear on LOW rather than HIGH. Overlap, length, and genre remain confounded, so these observations reject a simple “overlap causes Δ ” slogan without identifying an alternative single cause. A responsible scientific statement is therefore comparative and negative with respect to that slogan: high overlap is not required for positive coverage Δ in this design.

10.4 Length and genre as disclosed confounds

UNSEEN’s smaller matched-six mean (+5.78 pp) should not be narrated as “weaker priming because the domain is unseen” without acknowledging that UNSEEN is also shorter and expository. Likewise, HIGH’s mixture of large positives and Gemini negatives cannot be attributed solely to overlap. The cross-regime section therefore prioritizes pattern description over monocausal storytelling.

10.5 RQ4

Configuration heterogeneity is first-order: sign reversals, attenuation, amplification, and invalid cells coexist in the same experimental frame. Collapsing the study into a single universal priming effect would misrepresent the evidence.

10.6 Practical reading for manuscript users

For readers who need a one-paragraph takeaway without collapsing heterogeneity, the safest summary is: matched-six mean Δ is positive in HIGH, LOW, and UNSEEN; UNSEEN replicate Δ s are uniformly positive; HIGH uniquely contains negative Gemini means; LOW uniquely lacks a valid Qwen paired estimate; qualitative audits show that HIGH positives often involve opening reuse while LOW/UNSEEN positives often do not. Any shorter slogan loses at least one of those facts.

11 What the Qualitative Evidence Suggests About Mechanisms

We discuss mechanisms cautiously as hypotheses constrained by audits, not as proven causal pathways. The goal is to connect RQ5–RQ6 to the quantitative tables without over-fitting a single story to every configuration.

Hypothesis A: lexical/register calibration. Priming may increase the density of corpus-consistent ISV lexical choices and reduce Polish-residue forms, moving canonical coverage without rewriting source structure. LOW and UNSEEN positives with retained plot/exposition are compatible with this hypothesis. Under this view, the corpus functions less as a narrative template and more as a register/lexicon prior exposed in context.

Hypothesis B: orthographic normalization. Outside-inventory reductions often co-occur with positive Δ , suggesting that orthographic cleanup is one pathway by which the metric can move. Divergent cases show it is neither necessary nor sufficient. Metric–ortho divergence is itself scientifically useful: it warns against treating any single diagnostic as a complete account of Primed behaviour.

Hypothesis C: corpus-opening reuse under high overlap. On HIGH, large positives frequently co-occur with winter-opening near-copy. This may inflate coverage when corpus lexis enters the output, but it is not the only route to positive Δ , and opening copy can appear with negative Δ (Qwen HIGH r03). Hypothesis C therefore explains a HIGH-local signature rather than the whole experiment.

Compatibility, not competition. A, B, and C are not mutually exclusive. A configuration can show orthographic cleanup and lexical borrowing; a HIGH run can show opening reuse plus lexical adoption. The important empirical constraint is that A/B-compatible behaviour on LOW/UNSEEN demonstrates that C is not required for positive coverage Δ .

What we do not claim. We do not claim that models “learned Interslavic” from the corpus in a demonstrated causal sense, that priming is the sole cause of Δ , or that metric gains equal communicative quality. A useful descriptive framing—not a formal law—is that priming appears to be a *context-sensitive intervention* whose observed effect depends on the interaction between model configuration and source regime.

Consequences for future experiments. If Hypothesis C dominates in a future high-overlap narrative arm, experiments should measure opening overlap explicitly rather than treating it as anecdotal. If Hypotheses A/B dominate in low-overlap or unseen arms, evaluation should track lexical inventory hits and orthography separately from narrative similarity. Phase 2B already separates those diagnostics; future designs can pre-register which diagnostic is primary for which regime.

12 Reproducibility and Operational Findings

12.1 Qwen LOW invalid pairing

For LOW Qwen 3.8 Max — Fast, Direct runs used the intended model, but Primed runs were executed with **Qwen3.7-Plus** (default-selected in the chat interface). The paired cell is therefore `INVALID -- MODEL MISMATCH`. Any historical numerical Δ for that cell (including -12.08 pp) must not be interpreted as a Qwen 3.8 Max LOW priming effect. A later attempt to run the intended Qwen3.8-Max Primed condition encountered a service/security block; no corpus-shortening or prompt-evasion workaround was performed. Outputs were retained with validity sidecars. Incident record: `experiments/exp004-modelscreen/phase2b/a`

12.2 Gemini multi-message Primed protocol

Gemini Primed runs deliver the full corpus in message 1 and the translation request in message 2. Corpus bytes are not shortened. This is a recorded interface/context-window constraint applied consistently across regimes, not a redesign of the priming material.

12.3 Artifact provenance

Canonical experiment root: `experiments/exp004-modelscreen/phase2b/`. Final synthesis: `analysis/combined/`. Historical checkpoint files that predate UNSEEN completion are not used as the current experimental state for manuscript claims.

12.4 Replication package expectations

A reader seeking to regenerate the quantitative tables should start from the regime analysis JSON files and the combined synthesis, not from historical checkpoint prose. The manuscript’s `EVIDENCE_MAP.md` lists stable paths for each major claim. Raw outputs may remain local under copyright and repository policy; the committed evidence documents are sufficient to verify the numbers reported here.

12.5 What changed after HIGH+LOW

UNSEEN execution and analysis completed after the historical HIGH+LOW checkpoint. The final combined synthesis therefore supersedes any manuscript language that would treat UNSEEN as pending. Readers comparing older repository snapshots to this PDF should use artifact dates and the combined evidence pack rather than checkpoint prose alone.

13 Limitations

1. **Small repeat count.** Each configuration cell uses $n=3$ repeats; analysis is descriptive only.
2. **No inferential testing.** The design does not support significance claims or population generalization.
3. **Invalid Qwen LOW cell.** Same-model LOW priming for Qwen 3.8 Max is not estimated.
4. **Gemini protocol constraint.** Multi-message Primed delivery differs from single-message packaging used elsewhere.

5. **Source length differences.** HIGH $\gg$ LOW > UNSEEN in bytes; magnitude comparability is limited.
6. **Genre differences.** HIGH/LOW are narrative; UNSEEN is expository/technical.
7. **UNSEEN provenance.** Exact bibliography of the teaching-derived source is unknown.
8. **Metric sensitivity.** Canonical coverage can move with lexical/orthographic changes that are not full quality gains.
9. **No causal identification.** The paired contrast does not isolate corpus priming from all other session factors.

We do not invent additional limitations beyond those supported by the project artifacts.

These limitations jointly imply that Phase 2B is best read as a carefully documented observational contrast under controlled prompts, not as a definitive assay of Interslavic translation quality or of LLM linguistic competence.

14 Discussion

14.1 What the experiment establishes

Across three source regimes, authentic ISV corpus priming is associated with generally positive but strongly configuration-dependent shifts in canonical coverage. Matched-six mean Δ is positive on HIGH, LOW, and UNSEEN. UNSEEN shows uniform replicate positivity. Qualitative audits show that positive Δ can occur with or without corpus-opening near-copy, depending on regime. Operationally, the study also establishes that execution integrity (same-model pairing; recorded protocol deviations) must be part of the scientific record, because one invalid cell and one multi-message constraint materially affect interpretation.

14.2 What it does not establish

The study does not establish that priming always improves translation quality, that overlap causes larger effects, that any model is best, or that observed Δ is causally produced by corpus content alone. It also does not establish statistical significance. Readers should resist converting matched-six means into a claim that “priming improves ISV by about 6–9 points” as if that were a portable constant.

14.3 Why cross-regime consistency matters

If positive Δ appeared only on HIGH, a plausible scepticism would be that results merely reward thematic overlap. LOW and UNSEEN reduce that scepticism for the coverage metric, while still leaving length/genre confounds. Cross-regime persistence is therefore evidence of broader association, not proof of a universal intervention. In particular, UNSEEN matters because it extends the association beyond narrative fiction into a short educational scientific domain.

14.4 Why configuration heterogeneity matters

Gemini’s HIGH→LOW reversal shows that regime-averaged slogans fail. Engineering or scientific summaries that report only a global mean would hide opposite behaviours that are empirically present. Heterogeneity also cautions against treating vendor labels as interchangeable “LLM translation” units: the same priming corpus and source can yield different signed outcomes under different configurations.

14.5 Metric vs. quality

Because coverage is sensitive to inventory hits and orthography, Discussion must keep RQ5–RQ6 in view: textual mechanisms that move the metric are only partially aligned with naive notions of “better translation.” A Primed output can gain coverage by adopting inventory-supported forms while still containing naming instability, formatting artifacts, or unassessed adequacy errors. Future work should add human evaluation protocols before claiming quality gains.

14.6 A restrained conceptual framing

A useful descriptive framing—not a formal causal law—is that priming appears to be a context-sensitive intervention whose observed effect depends on the interaction between model configuration and source regime. That framing fits the tables without requiring an overfitted story about overlap, length, or a single textual mechanism.

14.7 Implications for interpreting “priming success”

If “success” is defined as a positive configuration mean Δ on a single high-overlap narrative, several configurations look successful on HIGH while Gemini does not. If “success” is defined as persistence of positive mean Δ on LOW and UNSEEN, Gemini looks successful there while the HIGH negatives remain part of the same evidence pack. The only non-misleading definition for this paper is therefore regime-conditional and configuration-conditional. That definition is less slogan-friendly, but it matches the tables.

14.8 Relation to in-context learning literature

Prior LLM-MT work already shows that prompt content and domain matter [Hendy et al., 2023, Moslem et al., 2023, Zhu et al., 2024]. Our contribution is not to rediscover that fact in the abstract, but to measure a specific Direct/Primed contrast for Polish-to-ISV under three deliberately chosen source regimes with frozen artifacts and paired repeats. The ISV coverage metric then makes the result auditable inside this repository even though it is not a substitute for human quality evaluation.

14.9 Answering RQ4–RQ6 in discussion form

RQ4 is answered by heterogeneity itself: reversals, attenuation, missing cells, and stable positives coexist. RQ5 is answered by the audit inventory: opening reuse, lexical borrowing, orthographic cleanup, naming churn, and structure preservation. RQ6 is answered negatively with respect to any single necessary textual mechanism: corpus-opening copying is salient on HIGH but not required for positive Δ on LOW/UNSEEN.

14.10 External validity

The study uses contemporary vendor interfaces and named configurations recorded at collection time. External validity to other interfaces, later model snapshots, or open-weights decoding stacks is unknown. Internal validity for the reported paired contrasts depends on the frozen sources, frozen corpus hash, and documented invalidation of the mismatched Qwen LOW cell.

15 Implications for Low-Resource Translation Systems

Future engineering direction (not an experimental result). A practical Polish-to-ISV assistant could, in principle, generate both Direct and Primed candidates, score them with deterministic resource-supported metrics, and expose disagreements for human review rather than assuming Primed is always preferable. Gemini’s HIGH negatives illustrate why blind Primed-only routing would be unsafe under the present evidence.

What is not validated here. We do not present a production translator architecture, latency study, user study, or deployment evaluation. Any such system remains future work. The present contribution is an evidence pack for when corpus priming moves a specific coverage metric under controlled regimes.

15.1 Evaluation-in-the-loop as a research agenda

Beyond candidate comparison, the results motivate evaluation-in-the-loop research for under-resourced targets: deterministic resource-supported metrics can flag large Primed/Direct disagreements for human inspection. That agenda is complementary to learned metrics such as COMET [Rei et al., 2020] and LLM-based evaluation [Kocmi and Federmann, 2023], which remain future options once ISV-compatible references or human ratings are available.

16 Conclusion

We reported a controlled descriptive study of authentic ISV corpus priming for Polish-to-ISV LLM translation across HIGH, LOW, and UNSEEN source regimes. Matched-six mean canonical-coverage Δ was +7.39 pp (HIGH), +8.53 pp (LOW), and +5.78 pp (UNSEEN). Effects were heterogeneous across configurations, including Gemini sign reversals on HIGH versus LOW/UNSEEN. Qualitative audits linked some HIGH positives to corpus-opening reuse, while LOW/UNSEEN positives often showed lexical and orthographic shifts without winter-narrative copying. One Qwen LOW pairing was invalid and excluded. Within these limits— $n=3$, descriptive analysis, coverage rather than full quality—the evidence supports a restrained claim: corpus priming is associated with generally positive but configuration-dependent coverage shifts that can persist beyond high-overlap narrative sources.

Future revisions may add human evaluation, additional unseen-domain sources with orthogonalized length, and verified Interslavic language-description citations. Those extensions would strengthen quality and linguistic framing; they are not required to interpret the descriptive coverage contrasts already measured here.

Declarations and Research Transparency

Funding. [PLACEHOLDER: funding not reported / none currently identified; confirm before submission.]

Competing interests. [PLACEHOLDER: competing interests not yet formally confirmed; confirm before submission.]

Data availability. Experiment protocols, frozen hashes, analysis artifacts, and the cross-regime synthesis are available in the project repository under `experiments/exp004-modelscreen/phase2b/` (canonical) and the curated mirror `docs/research/exp004-phase2b/`. Raw model outputs and copyrighted source texts may remain local/gitignored under repository policy; machine-readable aggregates and evidence documents needed to reconstruct manuscript tables are cited in `EVIDENCE_MAP.md`.

Ethics statement. No human participants or animals were involved in the experimental procedures reported here. The study evaluates LLM translation outputs on frozen text sources.

AI-use disclosure. This manuscript was drafted with assistance from LLM-based writing tools. Human authors remain responsible for scientific content, for verification of all numerical claims against canonical analysis artifacts, and for verification of bibliographic entries. AI systems are not authors.

Author contributions. Grzegorz R. Kulesza: conceptualization, experimental design/operation, analysis oversight, manuscript drafting and verification. [PLACEHOLDER: additional author contributions pending confirmation; see `AUTHORSHIP_PENDING.md`.]

Appendix A: Extended descriptive notes

This appendix restates selected descriptive facts already present in the combined synthesis so that the main text can remain readable while the PDF remains self-contained for review.

A.1 Matched-six membership

The matched-six comparison includes Gemini ON, Gemini OFF, Claude Sonnet 5, DeepSeek Expert ON, GPT-5.6 Luna, and Grok 4.5 Fast. It excludes Qwen 3.8 Max Fast solely because the LOW paired cell is invalid. HIGH and UNSEEN Qwen results remain valid within their regimes and are reported in Table 3, but they are not averaged into the matched-six means.

A.2 UNSEEN uniform positivity

UNSEEN contributes 21 paired replicate Δ values, all positive, ranging from +1.27 pp (Qwen r02) to +9.62 pp (Grok r02) in the combined synthesis. Uniform positivity is a sample property of these 21 pairs; it is not a significance claim and does not imply that every future unseen-domain source would behave identically.

A.3 HIGH mixed-sign cells

On HIGH, mixed-sign replicate patterns appear for Gemini ON (+1.89, −3.77, +0.23 pp), Gemini OFF (−5.39, +3.12, −2.98 pp), and Qwen (+8.70, +8.07, −1.23 pp). These cells illustrate why mean Δ alone can mislead when signs disagree within $n=3$.

A.4 LOW Grok mixed cell

On LOW, Grok is the only valid configuration with a mixed-sign pattern among primary cells; one replicate is negative (−1.50 pp) while the configuration mean remains positive (+3.95 pp). This is a descriptive caveat for LOW Grok, not a reason to invalidate the configuration.

A.5 Artifact precedence

If a historical checkpoint states that UNSEEN is incomplete, the current UNSEEN analysis and analysis/combined/synthesis take precedence for manuscript purposes. Historical HIGH↔LOW-only synthesis remains useful as a dated exploratory record but is not the final three-regime evidence pack.

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